Doomscroll Breaker: where to point in the code

For each requirement, open the file and scroll to the line numbers shown. Line numbers match the code at commit `bc29d37`.

MCX `mcxc444/MCXC444_Project/source/MCXC444_Project.c` (open in MCUXpresso)
ESP `doomscroll_esp32_wifi/doomscroll_esp32_wifi.ino` (on-board Wi-Fi version, the one to demo)
CFG `mcxc444/MCXC444_Project/freertos/freertos-kernel/template/FreeRTOSConfig.h`
HLP `tools/doomscroll_keys.py` **EXT** `extension/gate.js`, `extension/background.js`

Sensors and actuators (5 marks)

Requirement	Show
4 sensors	**MCX** `initJoystick` 294, `initInputs` 308 (button + sound sensor); **ESP** `initEncoder` 356 (EC11).
3 actuators	**MCX** `initLCD` 243 (`lcdShow` 273, `lcdNumber` 283), `initLED` 189 / `setLED` 207, `initBuzzer` 212 / `buzzerTone` 224.
≥ 1 sensor on MCXC444; max 1 on-board sensor and actuator	**MCX** pin `#define`s 23–32: 3 external sensors on the MCXC444. Only the LCD is on-board; no on-board sensor used.
Why each board (justification)	Joystick, button and sound sit on the MCXC444, which runs the session logic. The EC11 sits on the **ESP** because the ESP32 sends the volume keys over USB HID (`pollEncoder` 371, `pressVolume` 365).

Polling and interrupts (2 marks)

Polled sensor	**MCX** `joystickTask` 564–579: reads the ADC (`readJoystick` 301) every 20 ms via `vTaskDelay` 577. Also **ESP** encoder push switch, polled and debounced in `pollEncoder`.
Interrupt sensor	**MCX** `initInputs` 315–322 sets IRQC to either edge and enables PORTA in the NVIC; handled in `PORTA_IRQHandler` 325. NVIC levels: PORTA 1, UART2 2 (47–48). Also **ESP** `encoderIsr` 349, attached at 361–362.

ISR design (3 marks)

Button / sound ISR	**MCX** `PORTA_IRQHandler` 325–350: only reads the tick, debounces, then `xQueueSendFromISR` (337, 345) and `portYIELD_FROM_ISR` 349.
UART ISR, never waits for input	**MCX** `UART2_FLEXIO_IRQHandler` 372–396: handles one byte per interrupt. TX: next byte, then `xSemaphoreGiveFromISR(txDoneSem)` 382. RX: collects a 5-byte frame and hands it over with `xQueueSendFromISR(rxQueue)` 391. The checksum check is in a task (641).

Task design (4 marks)

≥ 3 tasks, with priorities	**MCX** `main` 706–711: 6 `xTaskCreate` calls, priorities 4 (uartRx), 3 (control), 2 (joystick, buzzer, uartTx), 1 (display). Scheduler starts at 712. **ESP** also runs `netTask` (699) beside `loop`.
Tasks yield	Each task blocks: `controlTask` on its queue 527 (1 s timeout), `joystickTask` delay 577, `buzzerTask` 586/589, `displayTask` 631, `uartRxTask` 640, `uartTxTask` 668/676.

Synchronisation tools (2 marks)

Mutex / semaphores	**MCX** created 692–693. `sessionMutex` in `controlTask` 528/560 and `displayTask` 599–601. Binary semaphore `txDoneSem`: given in ISR 382 → taken by `uartTxTask` 676.
Queues	**MCX** created 688–691. `eventQueue` fed at 337, 345 (ISR), 572 (joystick), 659 (uartRx) and read at 527; `soundQueue` 404 → 586; `txQueue` 400 → 668; `rxQueue` 391 (ISR) → 640.
Shared data with an ISR	**MCX** RX frames are copied into `rxQueue` by value (391), so no buffer is shared. TX: `txFrame`/`txIdx` are `volatile` (166–167) and filled in `uartTxTask` 669–674 before the TX interrupt is enabled at 675.

On the ESP32 too	**ESP** `uartMutex` (created 687) around `Serial1.write` 99–101, since `loop` and `netTask` both send packets; `noInterrupts` in `pollEncoder` 372–376; `netQueue` created 688, sent at 122 (Telegram ask jumps to the front), received at 656.

RTOS enabled (2 marks)

Pre-emption and time slicing	**CFG** lines 54–57: `#undef` then `#define configUSE_PREEMPTION 1` and `configUSE_TIME_SLICING 1`. These override the SDK's generated file, which set them to 0.

ESP32 use (7 marks)

Libraries beyond the template	**ESP** includes 14–22: WiFi, WiFiClientSecure, HTTPClient, ArduinoJson, Preferences, USB, USBHIDKeyboard, USBHIDConsumerControl.
Meaningful work	**ESP** USB keyboard `pressKey` 200; volume knob `pollEncoder` 371; budget saved in flash `saveBudget` 130 / `loadBudget` 152; midnight reset `applyDate` 275 fed by SNTP in `checkClockDate` 411.
Main loop and task	**ESP** `loop` 703–710; network task `netTask` 625, created at 699.

Communication protocol (5 marks)

UART setup	**MCX** `initUART2` 352–370 at `BAUD_RATE` 9600 (45). **ESP** `Serial1.begin(9600, …)` 686 on pins 26–27.
Packetisation	Packet types **MCX** 54–65, **ESP** 46–57. Build: **MCX** `uartTxTask` 669–673, **ESP** `sendPacket` 95–102. Parse and check XOR: **MCX** ISR framing 387–392 + `uartRxTask` 641, **ESP** `readMcx` 254 (XOR 266).
Two-way exchange	MCX → ESP: **ESP** `handlePacket` 205 (NEXT presses a key 219–220, STATE saves the budget 236, ASK queues the Telegram job 246–248). ESP → MCX: SYNC 162, APPROVED / DENIED 566 (Telegram), 296–298 (helper), 671 (timeout); handled in **MCX** 646–655.
Control flow (handshake)	**MCX** sends HELLO at boot 704 → **ESP** answers with SYNC 208–211 → **MCX** `onSync` 490.

Online functions (5 marks)

Telegram approval	**ESP** HTTPS in `telegramGet` 445; `askFriend` 506 (inline Approve/Deny buttons), `checkFriendReply` 536 (ArduinoJson filter 544–550, `answerCallbackQuery` 563). Laptop path: **HLP** `ask_friend` 137.
Firebase database	**ESP** `firebaseWrite` 578, called from `runJob` 597 (POST /sessions 614, PUT /budget 618); jobs come from `endSession` 183 and `uploadBudget` 166. Laptop path: **HLP** `firebase` 179, used at 206 and 212.
Wi-Fi and fallback	**ESP** `netTask`: `WiFi.begin` 628, SNTP `configTime` 644, reconnect 650. If an online call fails, `handToHelper` 106 prints `$ASK` / `$SESSION` / `$BUDGET` (109–115) for the helper.

State machine and extras

State machine	**MCX** `controlTask` 521–562 calls `onShortPress` 412, `onLongPress` 439, `onNext` 451, `onSync` 490, `onFriendReply` 499; `checkAskTimeout` 514. The state is sent on change at 557–558.
LCD / LED per state	**MCX** `displayTask` 595–633: amber at ≤ 3 on 611–612, “dEny” for 3 s 620–621, blinking red while waiting 627.
Only new videos count	**MCX** `onNext` 459–464 (`pos` vs `maxPos`).
Browser lock	**ESP** `printState` 332 → **HLP** `LockHandler` 60 → **EXT** `background.js` `readState` 8 → `gate.js` `check()` 82.